

When Marginal Shift Magnitudes Cannot Identify Conformal Coverage Loss

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The same drift magnitude can mean different coverage

Equal source-to-target discrepancy values can discard the direction that matters to the predictor.

Aligned translation

The shift follows the model's score direction. Scores change, and prediction-set coverage can change.

Orthogonal translation

The equally large shift is perpendicular to the score direction. Scores and coverage stay unchanged.

What the monitor sees

Rotation-invariant drift magnitudes agree. Raw target samples still retain direction and can distinguish the two laws.

The harmful gap can approach 40 percentage points

Gaussian construction with source error 10%, $b = \sqrt{\tau}$; orthogonal excess error is always zero.

Source scale τ	Absolute shift b	Aligned excess error
1.00	1.000	1.74 percentage points
0.50	0.707	8.46 percentage points
0.25	0.500	18.83 percentage points
0.10	0.316	30.90 percentage points
0.01	0.100	37.50 percentage points

- The theorem's limiting gap is 40 points; standardized shift diverges as $\tau \rightarrow 0$.

With a known label law, confidence is enough

Assume $P(Y = 1 \mid X) = \sigma(s(X))$ in source and target; let $T = |s(X)|$.

$$\theta_q(Q) = E_Q[\sigma(-T) \cdot 1\{T > q\}], \quad q \geq 0$$

- θ_q : target miscoverage; q : set threshold; σ : logistic function; the displayed formula uses $q \geq 0$.
- The equivalence between absolute-score laws and complete coverage curves uses all real thresholds.
- This requires the known link, not source calibration alone.

Perfect source calibration can still hide target error

A constructed covariate-shift example: $\Gamma = 2$, unchanged full signed-score distribution.

Distribution	Confidence for every case	Miscoverage
Source	90%	10%
Target: upweight hidden error group	90%	20%
Target: remove hidden error group	90%	0%

- Applies for $0 \leq q < \log 9$; the full conditional label law stays unchanged.

Without the link, compute a sharp interval

From compressed source calibration, target confidence frequencies, and a specified density-ratio cap Γ .

$$\max\{0, a - \Gamma(1 - m)\} \leq E_P[\omega e \mid T] \leq \min\{a, \Gamma m\}$$

- $e(X)$: error probability given full covariates; $m(T)$: source average of e at magnitude T .
- $a(T)$: target/source magnitude-frequency ratio; ω : covariate density ratio.
- The middle term is error mass per unit source magnitude mass, not target conditional error.
- Integrating over source magnitudes above q is sharp for these compressed quantities; fixed full source kernels can yield tighter bounds.

Finite samples produce simultaneous sensitivity bands

Source labels + unlabeled target scores + a specified cap $\rightarrow$ a coverage sensitivity interval.

- Freeze the score, confidence bins, and candidate threshold grid.
- Estimate source correct/error masses and target bin frequencies.
- Add simultaneous concentration slack before applying the sharp population bounds.
- Choose only thresholds whose upper error bound meets the target; Γ remains an assumption.

Drift detection and twin discrimination are different

Simulation: 80 repetitions, 160 observations/group, 199 permutations, fixed bandwidth.

Comparison	Representation	Rejections / 80
Source / aligned target	Raw features	80
Source / orthogonal target	Raw features	80
Aligned / orthogonal targets	Raw features	80
Aligned / orthogonal targets	Radius	6
Source / aligned target	Frozen score	80
Source / orthogonal target	Frozen score	6

- Wilson 95% intervals: $80/80 \rightarrow [0.95, 1.00]$; $6/80 \rightarrow [0.03, 0.15]$.

Learned-score replay supports intervals, not point claims

Controlled replay on two UCI datasets; 72 conditions and 100 replays per condition.

Reported quantity	Result	Interpretation
Simultaneous band containment	7,200 / 7,200	Under the imposed replay model
Worst source-bin point bias	0.0174	Unchanged bins can hide error change
Conditions improving risk-only bounds	9 / 72	Refinement is not universal
Largest population-bound tightening	0.1661	Finite intervals remain conservative

Information requirements for coverage monitoring

Match the conclusion to the information retained and the assumptions supplied.

- Drift magnitude: a change alarm can leave coverage loss unidentified.
- Known binary link: confidence identifies the entire coverage curve.
- Source calibration + ratio cap: report an identified interval.
- Unchanged confidence alone is not a target coverage certificate.

Source: github.com/ChorokLeeDev/conformal-coverage-identification-aistats2027